

Introduction to Lists in Python

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Introduction to Python Lists

- ▶ A list is a *collection which is ordered and changeable.
- ▶ Lists are containers that store data of any data type
- ▶ Lists allow duplicate members.
- ▶ Lists are written with square brackets.

**Collection is a data structure that can hold multiple values in a single variable*

Creating a List

- ▶ Example:

```
thislist = ["apple", "banana", "cherry"]
```

```
print(thislist)
```

Output: ['apple', 'banana', 'cherry']

- ▶ Use `type()` to check the type of list

Using the List Constructor

```
thislist = list(("apple", "banana", "cherry"))  
print(thislist)
```

Accessing List Items

- ▶ List items are indexed, the first item has index [0].

- ▶ Example:

```
thislist = ["apple", "banana", "cherry"]
```

```
print(thislist[1])
```

Output: banana

- ▶ Lists can store items of any data type

```
newlist=["Banana",50,True,15.36]
```

Negative Indexing

- ▶ Negative indexing means start from the end.
- ▶ -1 refers to the last item, -2 refers to the second last item.
- ▶ Example:

```
thislist = ["apple", "banana", "cherry"]
```

```
print(thislist[-1])
```

Output: cherry

Range of Indexes

- ▶ Specify a range of indexes to get multiple elements.

- ▶ Example:

```
thislist = ["apple", "banana", "cherry", "orange",  
"kiwi", "melon", "mango"]
```

```
print(thislist[2:5])
```

Output: ['cherry', 'orange', 'kiwi']

Changing Item Value

- ▶ To change the value of a specific item, refer to the index number.
- ▶ Example:

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist[1] = "blackcurrant"
```

```
print(thislist)
```

```
Output: ['apple', 'blackcurrant', 'cherry']
```


List Length

- ▶ Use the `len()` function to determine the number of items in a list.

- ▶ Example:

```
thislist = ["apple", "banana", "cherry"]
```

```
print(len(thislist))
```

Output: 3

Adding Items

- ▶ Use `append()` to add an item at the end of the list.
- ▶ Example:

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist.append("orange")
```

```
print(thislist)
```

Output: ['apple', 'banana', 'cherry', 'orange']

Inserting Items

- ▶ Use `insert()` to add an item at a specific index.
- ▶ Example:

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist.insert(1, "orange")
```

```
print(thislist)
```

Output: ['apple', 'orange', 'banana', 'cherry']

Using Extend()

- ▶ Use extend() to append elements from another list to current list
- ▶ Example:

```
thislist = ["apple", "banana", "orange"]  
newlist = ["mango", "pineapple", "kiwi"]  
thislist.extend(newlist)  
print(thislist)
```

Also you can use + operators to add two lists

```
a3=a1+a2
```

Removing Items

- ▶ Use `remove()` to delete a specific item.

- ▶ Example:

```
thislist = ["apple", "banana", "cherry"]
```

```
thislist.remove("banana")
```

```
print(thislist)
```

Output: ['apple', 'cherry']

- ▶ Use `clear()` to clear the list

Removing Items

- ▶ Use `pop()` to delete an item at specified index

- ▶ Example:

```
L1 = [1, 8, 7, 2, 21, 15, 18, 89]
```

```
L1.pop(2)
```

- ▶ Using `del` keyword to delete the list or element at specified index

```
del(L1) or del(L1[0])
```

Sorting the List

- ▶ Use `sort()` to sort a list

- ▶ Example:

```
L1 = [1, 8, 7, 2, 21, 15, 18, 89]
```

```
L1.sort()
```

```
L1.sort(reverse=True)
```

```
L1.reverse()
```

Copying a List

- ▶ Use `copy()` to copy a list

- ▶ Example:

```
a1 = [1, 8, 7, 2, 21, 15, 18, 89]
```

```
a2=a1.copy()
```

- ▶ Use `list()` also to copy a list

- ▶ Example:

```
a1 = [1, 8, 7, 2, 21, 15, 18, 89]
```

```
a2=list(a1)
```

```
a3 =a1[:] #Using slice operator
```


THANK YOU